SOFTWARE REQUIREMENTS SPECIFICATION

RSCI FYP Portal

A centralized, multi-domain Final Year Project management portal for the complete academic lifecycle

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Riphah School of Computing & Innovation

Document control

Document identification

Document title	RSCI FYP Portal SRS
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Document status

Baseline for review

This Software Requirements Specification (SRS) is a project-manager-grade baseline for design, estimation, implementation, testing, acceptance, and governance.

> **Governance note.** Values identified as proposed defaults are configurable and require Academic Committee ratification before operational use.

Revision history

Version	Date	Change	Owner
1.0	14 Oct 2026	Initial complete SRS baseline	RSCI FYP governance team

Approval record

FYP Coordinator / date	Head of Department / date	Academic Committee / date

Normative language

Shall denotes a mandatory requirement. **Should** denotes a recommended design objective. **May** denotes an optional capability. Requirement identifiers remain stable across revisions.

Table of contents

1. Introduction 5
1.1 Purpose; 1.2 Scope 5
1.3 Definitions; 1.4 References; 1.5 Overview 6
2. Overall description 7
2.1 Product perspective; 2.2 Product functions 7
2.3 User classes and characteristics 8
2.4 Operating environment; 2.5 Constraints 9
2.6 Assumptions and dependencies 9
2.7 Lifecycle and requirement conventions 10
3. System features 11
M1 Authentication & User Management 11
M2 Role-Based Access Control 11
M3 Academic Session & Batch Management 12
M4 Multi-Domain Project Catalogue 12
M5 Group Formation & Registration 13
M6 Proposal Submission & Approval 14
M7 Supervisor Allocation 15
M8 Meeting Catalogue & Logbook 16
M9 Milestone & Deliverable Tracking 17
M10 Document Repository & Version Control 18
M11 AI-Usage Declaration & Compliance 19
M12 Evaluation & Rubric Management 20
M13 Notifications & Communication 21
M14 Dashboards, Analytics & Reporting 21
M15 Audit Trail 21

Table of contents (continued)

4. External interface requirements 22
4.1 User interfaces 22
4.2 Hardware interfaces 22
4.3 Software interfaces 23
4.4 Communications interfaces 23
5. Non-functional requirements 24
5.1 Performance; 5.2 Security 24
5.3 Reliability; 5.4 Usability 25
5.5 Scalability; 5.6 Maintainability 25
5.7 Data, backup & disaster recovery 26
6. Evaluation framework 27
6.1 Configurable stage weightages 27
6.2 Stage rubrics E1-E5 28-32
6.3 Performance levels and adaptations 33
7. Framework catalogue 34
7.1 Ten FYP project types 34-38
7.2 Common obligations 39
7.3 Shared learning objectives 40
8. Data model overview 41
8.1 Key entities 41-42
8.2 Core relationships and integrity 43
Appendix A - Glossary 44
Appendix B - Permission matrix 45
Appendix C - Open issues requiring decision 46
Appendix D - Acceptance and traceability 47

> **Reading map.** Sections 1-5 define the product baseline. Section 6 is a normative proposal whose weightages remain configurable. Section 7 preserves the source framework catalogue. Sections 8 and the appendices support implementation and governance.

1. Introduction

1.1 Purpose

This SRS is the single source of truth for designing, building, verifying, deploying, and governing the RSCI FYP Portal. It specifies the complete Final Year Project (FYP) lifecycle across all ten project types defined by the RSCI FYP Framework v1.0, from session setup and group formation through proposal approval, supervision, evidence management, evaluation, publication, and archival.

1.2 Scope

The product is a centralized, mobile-responsive web portal serving students, faculty, governance roles, evaluators, examiners, and industry collaborators.

In scope: 15 modules

1. Authentication & users
2. Role-based access
3. Sessions & batches
4. Project catalogue
5. Groups & registration
6. Proposal workflow
7. Supervisor allocation
8. Meetings & logbook
9. Milestones & deliverables
10. Document repository
11. AI-usage compliance
12. Evaluation & rubrics
13. Notifications
14. Dashboards & reporting
15. Audit trail

Out of scope

- Building a plagiarism-detection engine; the portal integrates with a selected service through an API.
- Student fee assessment or payment processing.
- Replacing the institutional Learning Management System (LMS).

Boundary principle

The portal owns FYP-specific workflow, records, evidence, decisions, and reporting. External services remain authoritative for their own functions.

1. Introduction (continued)

1.3 Definitions, acronyms, and abbreviations

Term	Meaning
RSCI	Riphah School of Computing & Innovation
FYP	Final Year Project
SRS	Software Requirements Specification
SDD	System Design Document
RBAC	Role-Based Access Control
MVP	Minimum Viable Product
BMC	Business Model Canvas
KPI	Key Performance Indicator
HOD	Head of Department
Viva	Oral defense examination

1.4 References

[R1] Riphah School of Computing & Innovation, *FYP Framework v1.0 - Final Year Project Types for BS Computing Programs*, supplied source document.

1.5 Document overview

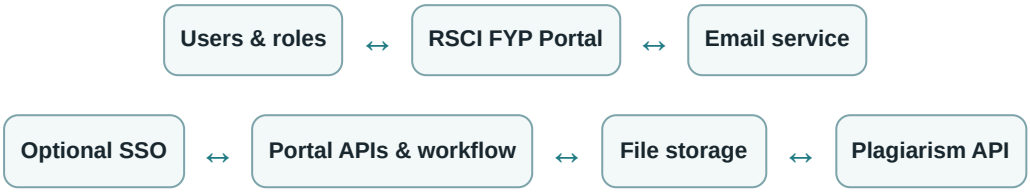
Section 2 describes actors, context, constraints, and assumptions. Section 3 defines functional requirements by module. Sections 4 and 5 define interfaces and quality attributes. Section 6 proposes configurable evaluation stages and rubrics. Section 7 encodes the framework catalogue and shared obligations. Section 8 outlines the domain model. Appendices provide terminology, permissions, open governance decisions, and acceptance guidance.

> **Authority rule.** Where this SRS conflicts with an approved later revision of the RSCI FYP Framework, the approved framework governs and the SRS shall be revised through change control.

2. Overall description

2.1 Product perspective

The RSCI FYP Portal is a centralized web application. It coordinates users and institutional workflows while delegating email delivery, file persistence, optional identity federation, and plagiarism analysis to controlled external services.



2.2 Product functions summary

- Establish academic sessions, programs, batches, calendars, and governance roles.
- Publish the ten-pathway catalogue with deliverable and AI-use rules.
- Form groups, submit structured proposals, and allocate supervision.
- Record meetings, actions, attendance, and effort evidence.
- Version deliverables and declarations against milestones.
- Configure panels, rubrics, scoring, moderation, and publication.
- Notify stakeholders and support group-level discussion.
- Provide dashboards, exports, archives, and immutable audit evidence.

> **Lifecycle principle.** Every material decision shall identify who acted, under which academic session and role, on which record version, at what time, and with what outcome.

2. Overall description - user classes

2.3 User classes and characteristics

User class	Primary responsibility	Key privileges
Super Admin	Platform operations	Global configuration; user provisioning; permission policy; integrations; audit access.
FYP Coordinator	Academic process owner	Session setup; catalogue; groups; allocations; deadlines; panels; moderation; reports.
HOD	Governance and oversight	Read-wide oversight; exception review; ratification visibility; published-result reports.
Supervisor	Primary academic mentor	Proposal review; meetings; feedback; deliverable acceptance; continuous assessment.
Co-Supervisor	Supporting mentor	Assigned-group access; feedback; meetings; contribution to permitted assessments.
Student (Leader/Member)	Project participant	Group membership; proposal and evidence submission; meetings; declarations; results.
Internal Evaluator	Institutional assessor	Assigned-stage evidence access; rubric scoring; comments; conflict declaration.
External Examiner	External assessor	Time-bound assigned evidence; rubric scoring; viva participation; comments.
Industry Partner	Client or mentor	Assigned-project requirements, meeting input, feedback, and acceptance evidence.

> **Separation of duties.** A user may hold multiple roles, but every action is authorized in the context of one academic session and one active role. Grade entry, moderation, and publication permissions shall be separable.

2. Overall description - environment and assumptions

2.4 Operating environment

- Standards-based modern browsers on desktop, tablet, and mobile devices.
- Responsive web UI over HTTPS, with server-side application services and a relational database.
- Institutionally approved object/file storage, SMTP-compatible email service, and optional SSO provider.
- Deployment may be on approved cloud or institutional infrastructure; environment choice shall not alter functional behavior.

2.5 Design and implementation constraints

- The system shall conform to the RSCI FYP Framework v1.0 until superseded through formal change control.
- The system shall follow applicable institutional data-privacy, record-retention, access, and academic-integrity rules.
- The product shall be web-based and mobile-responsive.
- Project type, program, group-size, load limits, workflow statuses, rubric weights, deadlines, and policies shall be configuration rather than hard-coded logic where specified.

2.6 Assumptions and dependencies

1. Evaluation weightages in Section 6 are proposed defaults pending Academic Committee ratification.
2. Group size defaults to 2-4 students and is configurable by session or program.
3. Maximum primary supervision load defaults to six active groups and is configurable.
4. Institutional email accounts and authoritative rosters are available.
5. RSCI identifies approved storage, email, and plagiarism providers before production deployment.
6. External examiners and industry partners complete approved onboarding and confidentiality steps.
7. Academic calendars and deadlines are supplied by authorized staff.
8. Data-retention and archival periods will be ratified before production go-live.

2. Overall description - lifecycle and conventions

2.7 Lifecycle states



Requirement conventions

Field	Interpretation
ID	Stable identifier: FR-[module]-[sequence] or NFR-[category]-[sequence].
Priority	Must = baseline release; Should = planned unless formally deferred.
Actor	The active user role or integration initiating the behavior.
Precondition	Authorization, session, or workflow state required before execution.
Acceptance evidence	Observable UI, API, database, notification, report, or audit result.

Cross-cutting business rules

- Every project record belongs to exactly one academic session, one program context, one project type, and one approved group.
- Published evaluations are read-only; corrections require an authorized, reasoned reopening and an auditable before/after record.
- Deadlines use the institution-configured timezone and retain both timestamp and timezone.
- Files are never overwritten in place; a new submission creates a new document version.
- Framework policies are versioned so historical projects retain the rule set under which they were approved.

3. System features

M1 Authentication & User Management

ID	Priority	Requirement
FR-UM-01	Must	The system shall authenticate users using verified institutional email addresses.
FR-UM-02	Must	The system shall enforce a configurable strong-password policy and account lockout controls.
FR-UM-03	Should	The system shall expose an optional SSO integration hook without removing local administrative recovery.
FR-UM-04	Must	The system shall maintain user profiles with identity, program/department, expertise, and contact fields appropriate to role.
FR-UM-05	Must	Authorized staff shall bulk import students and faculty by CSV per academic session and activate or deactivate accounts without deleting history.

M2 Role-Based Access Control

ID	Priority	Requirement
FR-RB-01	Must	The system shall enforce least-privilege permissions for all nine defined user classes on every protected request.
FR-RB-02	Must	The system shall scope each role assignment to an academic session and, where applicable, a program or project group.
FR-RB-03	Must	An authorized role owner shall delegate permitted duties for a defined period, scope, and reason.
FR-RB-04	Must	The system shall provide an auditable permission matrix and deny access by default when no explicit permission applies.

3. System features

M3 Academic Session & Batch Management

ID	Priority	Requirement
FR-AS-01	Must	The coordinator shall create, open, close, archive, and clone academic sessions such as Fall 2026.
FR-AS-02	Must	The system shall configure programs CS, SE, IT, AI, DS, CY, and CGV and associate intake batches with them.
FR-AS-03	Must	The coordinator shall publish an academic calendar containing milestones, deadlines, evaluation windows, and blackout dates.
FR-AS-04	Must	The system shall prevent cross-session data changes unless the user holds an explicitly scoped administrative permission.
FR-AS-05	Must	The system shall notify affected roles when an authorized calendar change moves a published deadline or evaluation window.

M4 Multi-Domain Project Catalogue

ID	Priority	Requirement
FR-PC-01	Must	Authorized staff shall create, read, update, retire, and version project-type records without changing historical projects.
FR-PC-02	Must	Each project type shall store suitable programs, description, objectives, outcomes, type-specific deliverables, and AI-usage policy.
FR-PC-03	Must	The baseline catalogue shall contain the ten project types listed in Section 7.
FR-PC-04	Must	Students shall search and filter active project types by program, domain, deliverable profile, and AI policy.
FR-PC-05	Must	Catalogue changes affecting active groups shall require a recorded effective date and coordinator acknowledgement.

3. System features

M5 Group Formation & Registration

ID	Priority	Requirement
FR-GR-01	Must	The coordinator shall configure minimum and maximum group size by session or program, defaulting to 2-4.
FR-GR-02	Must	Students shall create a proposed group, nominate one leader, and invite eligible members.
FR-GR-03	Must	Invitees shall accept or decline invitations, and the system shall prevent simultaneous active membership in multiple groups in one session.
FR-GR-04	Must	The coordinator shall approve, reject, or return a registration request with comments.
FR-GR-05	Must	On approval, the system shall issue a unique, immutable group ID and record subsequent membership changes with reasons.

3. System features

M6 Proposal Submission & Approval Workflow

ID	Priority	Requirement
FR-PR-01	Must	The proposal form shall capture title, project type, domain, abstract, problem statement, objectives, scope, methodology, tools/technologies, and timeline.
FR-PR-02	Must	The system shall validate required fields, group eligibility, and program suitability before submission.
FR-PR-03	Must	The workflow shall support Draft, Submitted, Supervisor Review, Coordinator Approval, Approved, and Rejected states.
FR-PR-04	Must	Supervisors and coordinators shall annotate, comment on, approve, reject, or return proposals according to their stage permissions.
FR-PR-05	Must	Students shall resubmit a revised proposal while preserving previous versions and review comments.
FR-PR-06	Must	Approval shall lock the accepted proposal baseline; later scope changes shall use a reasoned change request and approval trail.

3. System features

M7 Supervisor Allocation

ID	Priority	Requirement
FR-SA-01	Must	Faculty profiles shall include searchable expertise tags, availability, and current supervision load.
FR-SA-02	Must	Students shall submit ranked supervisor preferences within a configured window.
FR-SA-03	Must	The coordinator allocation console shall show preference, expertise match, conflicts, and load against a configurable default cap of six groups.
FR-SA-04	Must	The coordinator shall assign a primary supervisor, optional co-supervisor, and an industry mentor for applicable industry or consulting projects.
FR-SA-05	Must	Reassignment shall capture reason, effective date, approvals, retained history, and notifications to affected parties.

3. System features

M8 Meeting Catalogue & Logbook

ID	Priority	Requirement
FR-MT-01	Must	Authorized group participants shall schedule one-off or recurring supervisory meetings with date, time, venue/link, agenda, and invitees.
FR-MT-02	Must	Meeting records shall capture attendance, minutes, decisions, attachments, and action items with owner, due date, and status.
FR-MT-03	Must	A supervisor shall sign off, return, or correct submitted minutes, and the system shall retain the review history.
FR-MT-04	Must	Students shall log dated effort hours by member, activity category, and evidence reference; supervisors shall verify or query entries.
FR-MT-05	Must	The system shall roll up verified hours by student, group, week, and session toward 300 hours and display a 9-hours-per-week guideline indicator.
FR-MT-06	Must	Authorized users shall export a chronological, signed logbook including meetings, actions, attendance, and effort summaries.

3. System features

M9 Milestone & Deliverable Tracking

ID	Priority	Requirement
FR-DL-01	Must	The system shall generate a group checklist from the selected type-specific template plus the common obligations in Section 7.2.
FR-DL-02	Must	Each milestone shall define owner, deadline, required artifact, approval rule, and applicable evaluation stage.
FR-DL-03	Must	The submission workflow shall support Submit, Review, Revise, and Accept with timestamped comments and version references.
FR-DL-04	Must	The system shall flag late submissions using the applicable deadline and any authorized extension.
FR-DL-05	Must	Dashboards shall show completion, overdue, rejected, and at-risk milestones without treating a file upload as acceptance.

3. System features

M10 Document Repository & Version Control

ID	Priority	Requirement
FR-DR-01	Must	The repository shall store all submission versions with uploader, checksum, timestamp, file type, and linked workflow record.
FR-DR-02	Must	Authorized staff shall publish versioned templates including proposal, SRS, SDD, and final-report formats.
FR-DR-03	Must	Repository access shall be group-scoped and role-scoped, with time-limited access for external examiners and partners.
FR-DR-04	Must	The system shall create a read-only final archive per academic session with a manifest of projects and accepted artifacts.

3. System features

M11 AI-Usage Declaration & Compliance

ID	Priority	Requirement
FR-AI-01	Must	Every deliverable submission shall include a completed AI-usage declaration before it can enter review.
FR-AI-02	Must	Declaration options shall map to the active project-type policy: no AI in core work/documentation, API-only, extensive AI, or limited AI as applicable.
FR-AI-03	Must	The declaration shall capture tool or service, purpose, affected artifact sections, human verification, and student attestation when AI was used.
FR-AI-04	Must	The supervisor shall attest, question, or return the declaration during deliverable review.
FR-AI-05	Must	The system shall flag a policy mismatch for coordinator review without automatically determining misconduct.

3. System features

M12 Evaluation & Rubric Management

ID	Priority	Requirement
FR-EV-01	Must	Authorized staff shall configure evaluation stages, dates, stage weightages, eligibility gates, and publication rules by session.
FR-EV-02	Must	The rubric builder shall define criteria, maximum marks, weightage, performance-level descriptors, and type-specific variants.
FR-EV-03	Must	The coordinator shall assign evaluator panels, including internal evaluators and external examiners, and record conflicts of interest.
FR-EV-04	Must	Evaluators shall score each criterion within its permitted range and provide criterion or overall comments before submission.
FR-EV-05	Must	The system shall aggregate evaluator scores using configured rules and show calculation details to authorized moderators.
FR-EV-06	Must	Results shall move through Draft, Moderated, and Published states; publication shall require authorized approval.
FR-EV-07	Must	A post-publication correction shall require reopening authority, a reason, before/after values, re-moderation, and notification.

3. System features

M13 Notifications & Communication

ID	Priority	Requirement
FR-NC-01	Must	The system shall send in-app and email notifications for deadlines, meeting invitations/reminders, evaluation schedules, posted feedback, and published results.
FR-NC-02	Must	Users shall manage non-mandatory notification preferences while critical academic notices remain enabled.
FR-NC-03	Must	Authorized staff shall post targeted, time-bounded announcements by session, program, role, or group.
FR-NC-04	Must	Each group shall have a role-scoped discussion thread with attachments, timestamps, and moderation/audit controls.

M14 Dashboards, Analytics & Reporting

ID	Priority	Requirement
FR-AN-01	Must	Administrative dashboards shall show groups by type, at-risk groups, supervision load, deliverable status, and evaluation progress.
FR-AN-02	Must	Supervisor dashboards shall show assigned groups, upcoming meetings, overdue actions, pending reviews, and hours requiring verification.
FR-AN-03	Must	Student dashboards shall show timeline, progress, logged effort, pending actions, feedback, evaluation schedule, and published results.
FR-AN-04	Must	Authorized users shall filter and export agreed reports in CSV and PDF, and HOD users shall receive cross-program oversight views.

M15 Audit Trail

ID	Priority	Requirement
FR-AU-01	Must	The system shall maintain an immutable audit log of security, workflow, allocation, policy, evaluation, and publication actions.
FR-AU-02	Must	Each audit event shall store actor, active role, session, timestamp, action, target, outcome, and before/after values for grade changes.
FR-AU-03	Must	The system shall retain login history and allow authorized security review without exposing credentials or session secrets.

4. External interface requirements

4.1 User interfaces

ID	Interface requirement
UI-01	The UI shall be responsive from mobile through desktop widths and shall preserve all required actions without horizontal page scrolling.
UI-02	The target shall be WCAG 2.1 Level AA, including keyboard operation, visible focus, labels, contrast, error identification, and text alternatives.
UI-03	Role dashboards shall use the same navigation model while presenting role-relevant tasks, counts, deadlines, and exceptions.
UI-04	Forms shall provide field-level validation, save-draft behavior where applicable, confirmation for consequential actions, and a human-readable error summary.
UI-05	Dates, times, statuses, scores, and policies shall be displayed with their governing session context and without relying on color alone.

4.2 Hardware interfaces

No specialized end-user hardware is required. Users need a network-connected device capable of running a supported browser. For IoT project evidence, the portal stores documents, media, and verification records; it does not directly control student hardware.

Interaction patterns

Task-oriented

Pending reviews, actions, and deadlines appear before passive records.

Evidence-linked

Every decision links to the exact proposal, submission, declaration, or score version.

Role-transparent

The active role and session remain visible during privileged actions.

4. External interface requirements (continued)

4.3 Software interfaces

Interface	Purpose	Minimum contract
SMTP / email service	Transactional notifications	Authenticated delivery; templates; retries; delivery status; no grades in subject lines.
File/object storage	Versioned evidence	Upload/download; checksum; malware-scan hook; access control; retention; signed access where used.
Plagiarism-check API	Submit and retrieve similarity analysis	Provider selection configurable; request status; report link/reference; failure handling; consent and retention controls.
Optional SSO	Federated authentication	Standard institutional identity protocol; unique subject mapping; role provisioning remains portal-controlled.
CSV import/export	Roster and reporting interchange	Published schema; validation preview; row-level errors; UTF-8; import audit record.

4.4 Communications interfaces

- All browser and API traffic shall use HTTPS with current institution-approved TLS configuration.
- Service-to-service calls shall authenticate, time out, retry only safe operations, and record correlation identifiers.
- API payloads shall use documented, versioned formats and reject unexpected fields where they could affect authorization or grades.
- Email shall contain minimal sensitive data and direct authenticated users to the portal for protected content.

> **Integration constraint.** The plagiarism engine is outside scope. The portal records submission, provider status, and report reference; provider selection and academic interpretation remain institutional decisions.

5. Non-functional requirements

5.1 Performance and capacity

5.2 Security and privacy

ID	Priority	Requirement
NFR-PE-01	Must	For authenticated interactive pages under normal load, server-side response plus render-ready data shall meet a p95 target below 2 seconds, excluding large file transfer and third-party latency.
NFR-PE-02	Must	The production design shall support at least 500 concurrent authenticated users while meeting NFR-PE-01 in an agreed load test.
NFR-PE-03	Must	A 100 MB permitted upload shall show progress and shall not block unrelated requests; configurable size limits shall be enforced before storage finalization.
NFR-SE-01	Must	Security controls shall align with applicable OWASP ASVS practices selected during architecture review and recorded in the verification plan.
NFR-SE-02	Must	Authorization shall be enforced server-side for every protected object and action; hidden UI controls shall not be treated as access control.
NFR-SE-03	Must	Sensitive data shall be encrypted in transit and at rest using institution-approved mechanisms and keys.
NFR-SE-04	Must	Sessions shall use secure cookies or equivalent protected tokens, inactivity timeout, absolute lifetime, logout invalidation, and rotation after privilege change.
NFR-SE-05	Must	Grades, rubrics, moderation notes, and unpublished results shall be restricted to authorized roles and excluded from logs and notification previews.
NFR-SE-06	Must	The application shall validate uploads, prevent executable delivery, and provide a malware-scanning integration hook before files become downloadable.

5. Non-functional requirements (continued)

5.3 Reliability and availability

5.4 Usability and accessibility

5.5 Scalability

5.6 Maintainability and portability

ID	Priority	Requirement
NFR-RE-01	Must	The service shall target 99.5% monthly availability during active academic terms, excluding approved maintenance windows.
NFR-RE-02	Must	Workflow operations shall be transactional so a failed operation does not leave partial approvals, scores, or file references.
NFR-RE-03	Must	Notification and third-party failures shall be retryable without duplicating the underlying academic action.
NFR-US-01	Must	A first-time student shall be able to locate the active timeline, pending actions, and group record without training after login.
NFR-US-02	Must	The system shall provide consistent terminology, status explanations, actionable validation messages, and confirmation for irreversible or publication actions.
NFR-US-03	Must	At least the critical submission, review, meeting, and scoring workflows shall be keyboard-operable and tested against the WCAG 2.1 AA target.
NFR-SC-01	Must	The architecture shall support multiple concurrent and archived sessions, all listed programs, and catalogue growth without schema changes for each new type.
NFR-SC-02	Must	Reporting queries shall be isolated or optimized so they do not materially degrade submission and evaluation workflows.
NFR-MP-01	Must	Business rules for weightages, limits, deadlines, and policies shall be configuration with validation and version history.
NFR-MP-02	Must	The system shall expose documented deployment, migration, rollback, monitoring, and incident-diagnosis procedures.
NFR-MP-03	Must	The application shall avoid provider lock-in by isolating storage, email, SSO, and plagiarism integrations behind replaceable adapters.

5. Non-functional requirements - data protection

5.7 Data management, backup, and disaster recovery

ID	Priority	Requirement
NFR-DM-01	Must	The database shall enforce referential integrity and preserve immutable identifiers for users, groups, submissions, evaluations, and audit events.
NFR-DM-02	Must	Production backups shall run at least daily, be encrypted, monitored, and periodically restored in a non-production recovery test.
NFR-DM-03	Must	The proposed recovery point objective is 24 hours and the proposed recovery time objective is 8 hours; both targets require institutional approval and infrastructure validation.
NFR-DM-04	Must	Retention, legal hold, anonymization, archival, and disposal shall follow a ratified institutional schedule and produce auditable records.
NFR-DM-05	Must	Exports shall preserve the selected filter context, generation time, requesting user, and academic-session scope.
NFR-DM-06	Must	Audit records shall be append-only to application users and protected from alteration through privileged database access controls and monitoring.
NFR-DM-07	Must	Production data shall not be copied to development or test environments unless de-identified under an approved process.

Daily
Minimum backup frequency

24 h
Proposed RPO

8 h
Proposed RTO

Approval required. RPO, RTO, retention periods, and backup geography are proposed design targets, not confirmed RSCI policy. They shall be approved before production acceptance.

6. Evaluation framework

> **Normative proposal.** Every stage weightage on this page is a *configurable default pending Academic Committee ratification*. The portal shall not hard-code these values.

6.1 Proposed stage model

Stage	Name	Default weight	Purpose
E1	Proposal Defense	10%	Problem viability, plan, and readiness
E2	Mid-Term Evaluation	20%	Progress, technical depth, and documentation
E3	Continuous Supervisor Assessment	10%	Year-long discipline, ownership, and professionalism
E4	Final Demonstration	30%	Completed work, quality, validation, and demonstration
E5	Final Report & Oral Defense (Viva)	30%	Written evidence, analysis, and individual understanding

Calculation and governance rules

- Configured stage weightages for a session shall total 100% before evaluations can be published.
- Each stage rubric scores 100 raw points. The portal converts the moderated stage score by the ratified stage weight.
- Rubric version, scorer identities, aggregation rule, moderation decision, and published result shall remain traceable.
- Type-specific variants may reallocate criterion emphasis while retaining documented totals and approval history.
- Absent, deferred, excused, and misconduct-related outcomes require separate statuses; they shall not be represented as ordinary zero scores without an authorized decision.

6.2 Rubric - E1 Proposal Defense

> **Configurable default.** Criterion marks total 100 for this stage and remain subject to Academic Committee ratification.

E1 Proposal Defense

Criterion	Maximum marks
Problem definition & motivation	20
Objectives & scope clarity	15
Literature/background awareness	15
Methodology & work plan	20
Feasibility, timeline & risk awareness	15
Presentation & Q&A	15

Scoring controls

- Every score shall be within the criterion maximum and accompanied by the configured descriptor or comment requirement.
- Where multiple evaluators score independently, the configured aggregation rule shall be visible before submission.
- Moderators may return an evaluation for clarification; they may not silently edit an evaluator's submitted score.
- The system shall preserve draft, submitted, moderated, and published versions.

6.2 Rubric - E2 Mid-Term Evaluation

> **Configurable default.** Criterion marks total 100 for this stage and remain subject to Academic Committee ratification.

E2 Mid-Term Evaluation

Criterion	Maximum marks
Progress vs plan	25
Technical depth & quality of work	25
Documentation (SRS/SDD as applicable)	20
Teamwork & individual contribution	15
Presentation & Q&A	15

Scoring controls

- Every score shall be within the criterion maximum and accompanied by the configured descriptor or comment requirement.
- Where multiple evaluators score independently, the configured aggregation rule shall be visible before submission.
- Moderators may return an evaluation for clarification; they may not silently edit an evaluator's submitted score.
- The system shall preserve draft, submitted, moderated, and published versions.

6.2 Rubric - E3 Continuous Supervisor Assessment

> **Configurable default.** Criterion marks total 100 for this stage and remain subject to Academic Committee ratification.

E3 Continuous Supervisor Assessment

Criterion	Maximum marks
Regularity & meeting discipline	25
Initiative & professionalism	25
Technical ownership of progress	25
Documentation discipline	25

Scoring controls

- Every score shall be within the criterion maximum and accompanied by the configured descriptor or comment requirement.
- Where multiple evaluators score independently, the configured aggregation rule shall be visible before submission.
- Moderators may return an evaluation for clarification; they may not silently edit an evaluator's submitted score.
- The system shall preserve draft, submitted, moderated, and published versions.

6.2 Rubric - E4 Final Demonstration

> **Configurable default.** Criterion marks total 100 for this stage and remain subject to Academic Committee ratification.

E4 Final Demonstration

Criterion	Maximum marks
Completeness vs objectives	25
Technical complexity & quality	20
Innovation/novelty or business value (type-dependent)	15
Testing/validation & results	20
Demo & Q&A	20

Scoring controls

- Every score shall be within the criterion maximum and accompanied by the configured descriptor or comment requirement.
- Where multiple evaluators score independently, the configured aggregation rule shall be visible before submission.
- Moderators may return an evaluation for clarification; they may not silently edit an evaluator's submitted score.
- The system shall preserve draft, submitted, moderated, and published versions.

6.2 Rubric - E5 Final Report & Oral Defense (Viva)

> **Configurable default.** Criterion marks total 100 for this stage and remain subject to Academic Committee ratification.

E5 Final Report & Oral Defense (Viva)

Criterion	Maximum marks
Report structure & writing quality	20
Literature review	15
Methodology/design description	15
Results, analysis & evaluation	20
Conclusion & future work	10
Viva Q&A & individual understanding	20

Scoring controls

- Every score shall be within the criterion maximum and accompanied by the configured descriptor or comment requirement.
- Where multiple evaluators score independently, the configured aggregation rule shall be visible before submission.
- Moderators may return an evaluation for clarification; they may not silently edit an evaluator's submitted score.
- The system shall preserve draft, submitted, moderated, and published versions.

6. Evaluation framework - descriptors and adaptations

6.3 Performance-level descriptors

Level	Band	General descriptor
Excellent	85-100	Comprehensive, accurate, independent, and professionally evidenced performance.
Good	70-84	Strong performance with minor gaps that do not undermine the main outcome.
Satisfactory	55-69	Meets essential expectations but shows notable gaps in depth, consistency, or evidence.
Needs Improvement	40-54	Partially meets expectations; important weaknesses require remediation.
Unsatisfactory	Below 40	Does not demonstrate the minimum required competence or evidence.

6.4 Per-project-type rubric adaptations

Project type	Required adaptation
Research	Increase emphasis on novelty and experimental rigor; require a research paper draft.
Entrepreneurial	Emphasize MVP, business plan, BMC, financials, market evidence, and pitch.
Consulting - all four tracks	Emphasize analysis depth, recommendation quality, executive presentation, and a client-feedback component.
IoT & Embedded Systems	Require a live hardware prototype and hardware-software-cloud integration evidence.
Case Study	Emphasize analysis depth, root-cause reasoning, recommendations, and roadmap; no build is required.
Continuation / Industry	Require client acceptance report and deployment evidence.

> **Implementation rule.** A variant shall identify its parent rubric, changed criteria or marks, applicable project types, approving authority, effective session, and version. The portal shall reject a rubric whose criterion maximums do not total 100.

7. Framework catalogue - project types

> **Normative source.** Catalogue content on this page is derived from the RSCI FYP Framework v1.0. Portal records shall preserve policy version and effective session.

1. Product Development Project

Suitable programs: CS, SE, IT

Purpose: Complete SDLC-based software or hardware solution to a real-world problem.

Expected deliverables: Software application or computing system; source code repository; SRS; SDD; Test Plan and Test Report; User Manual; Technical Documentation; Final Project Report.

AI usage: No AI assistance in core functionalities or documentation; AI application through APIs is allowed.

2. Entrepreneurial Project

Suitable programs: CS, SE, IT

Purpose: Technology product or service with commercial potential, market validation, and an MVP.

Expected deliverables: MVP; Business Plan; Business Model Canvas; Market Research Report; Financial Feasibility Analysis; Pitch Deck; Product Demonstration.

AI usage: Extensive AI use is allowed for product development and documentation.

7. Framework catalogue - project types

> **Normative source.** Catalogue content on this page is derived from the RSCI FYP Framework v1.0. Portal records shall preserve policy version and effective session.

3. Continuation Project (Existing product based Project)

Suitable programs: CS, SE, IT, CY, DS, AI, CGV

Purpose: Industry-collaborative work solving an organizational problem using professional practice.

Expected deliverables: Industry-ready solution; Client Requirements Document; Deployment Plan; Client Acceptance Report; User Documentation; Final Project Report.

AI usage: AI may be allowed for new-feature development depending on the project; API-based AI is allowed; no AI use in documentation.

4. Research Project

Suitable programs: CS, AI, CY, DS

Purpose: Scientific investigation using literature, experimentation, data analysis, and validation.

Expected deliverables: Literature Review; Research Proposal; Experimental Design; Dataset where applicable; optional Prototype; Research Paper Draft; Final Research Report.

AI usage: No AI assistance in core functionalities or documentation; AI application through APIs is allowed.

7. Framework catalogue - project types

> **Normative source.** Catalogue content on this page is derived from the RSCI FYP Framework v1.0. Portal records shall preserve policy version and effective session.

5. IoT & Embedded Systems Project

Suitable programs: CS, SE, IT

Purpose: Integrated sensors, embedded devices, communications, cloud platforms, and applications.

Expected deliverables: Hardware Prototype; Embedded Software/Firmware; Cloud Dashboard; Mobile/Web Interface; Circuit Diagrams; Technical Documentation.

AI usage: No AI assistance in core functionalities or documentation; AI application through APIs is allowed.

6. Case Study Project

Suitable programs: IT, CY, DS

Purpose: Systematic investigation of an existing organization, technology, system, or computing problem.

Expected deliverables: Case Study Report; Current State Analysis; Root Cause Analysis; Recommendations; Implementation Roadmap.

AI usage: Limited AI assistance in core functionalities; no AI use in documentation.

7. Framework catalogue - project types

> **Normative source.** Catalogue content on this page is derived from the RSCI FYP Framework v1.0. Portal records shall preserve policy version and effective session.

7. Consulting Project (AI/Data Analytics)

Suitable programs: AI, DS

Purpose: Organizational data analysis and AI/analytics-driven recommendations for decisions and performance.

Expected deliverables: Business Problem Analysis; Data Analysis Report; AI/ML Models or Analytics Dashboard; Business Recommendations; Executive Presentation; Consulting Report.

AI usage: No AI assistance in documentation.

8. Consulting Project (Cybersecurity)

Suitable programs: CY, IT

Purpose: Security assessment, vulnerability analysis, risk management, compliance, and improvement advice.

Expected deliverables: Security Assessment Report; Vulnerability Assessment/Penetration Testing Report where applicable; Risk Assessment Matrix; Security Improvement Plan; Executive Summary; Consulting Report.

AI usage: No AI assistance in documentation.

7. Framework catalogue - project types

> **Normative source.** Catalogue content on this page is derived from the RSCI FYP Framework v1.0. Portal records shall preserve policy version and effective session.

9. Consulting Project (Cloud & DevOps)

Suitable programs: IT, CS, SE

Purpose: Cloud readiness, architecture, DevOps maturity, modernization, automation, and adoption strategy.

Expected deliverables: Cloud Readiness Assessment; Cloud Architecture Design; DevOps Maturity Assessment; CI/CD Pipeline Design; Migration Strategy; Consulting Report.

AI usage: No AI assistance in documentation.

10. Consulting Project (Digital Transformation)

Suitable programs: IT, CS, SE

Purpose: Digital maturity and process analysis leading to a transformation strategy and roadmap.

Expected deliverables: Digital Maturity Assessment; Business Process Analysis; Technology Roadmap; Digital Transformation Strategy; Cost-Benefit Analysis; Executive Presentation; Consulting Report.

AI usage: No AI assistance in documentation.

7. Framework catalogue - common obligations

7.2 Common framework obligations

The source framework numbers fifteen common obligations. Items 1-14 are artifacts or activities; item 15 is the required effort rule. The portal shall preserve this distinction while tracking all fifteen.

No.	Common obligation
1	Project Proposal
2	Problem Statement and Objectives
3	Literature Review / Background Study
4	Project Planning and Timeline
5	Requirements Analysis, where applicable
6	Design and Methodology
7	Implementation or Investigation
8	Validation, Testing, or Evaluation
9	Final Project Report
10	Poster and Demonstration
11	Oral Defense Presentation
12	Source Code, Prototype, Dataset, or Supporting Artifacts, as applicable
13	User and Technical Documentation
14	Reflection on Project Outcomes and Future Work
15	Project effort: 9 hours per week; 300 hours total per project

Portal enforcement

- The project-type checklist shall combine these common obligations with the type-specific deliverables in Section 7.1.
- “Where applicable” and “as applicable” items require a recorded applicability decision rather than silent deletion.
- Verified effort shall roll up by member and group against both the weekly guideline and total project target.

7. Framework catalogue - shared learning objectives

7.3 Learning objectives across all FYP types

ID	Learning objective
LO-1	Apply computing knowledge to solve complex real-world problems.
LO-2	Demonstrate technical competence in the chosen specialization.
LO-3	Integrate theoretical concepts with practical implementation.
LO-4	Conduct independent inquiry, analysis, and critical evaluation.
LO-5	Work effectively as individuals or within multidisciplinary teams.
LO-6	Communicate technical ideas professionally through written reports, presentations, and demonstrations.
LO-7	Follow ethical, legal, and professional standards in computing practice.
LO-8	Deliver solutions that create measurable academic, industrial, commercial, or societal impact.

Evidence mapping

Objective group	Indicative portal evidence
LO-1 to LO-4	Proposal, literature/background, design/methodology, implementation/investigation, validation, rubric scores.
LO-5	Membership, meeting attendance, action ownership, contribution records, supervisor assessment.
LO-6	Reports, poster, demonstration, presentation, viva rubric, feedback.
LO-7	AI declarations, ethics approvals where applicable, audit trail, professional-conduct evidence.
LO-8	Validation results, client acceptance, market evidence, recommendations, impact statement.

> **Assessment integrity.** Evidence mapping supports curriculum review and reporting; it does not replace evaluator judgment or automatically infer attainment from file submission.

8. Data model overview

8.1 Key entities - identity, context, and supervision

Entity	Purpose	Primary relationships
User	Institutional or invited identity	User 1..* RoleAssignment; User 0..* AuditLog
Role	Named access role	Role *..* Permission via assignments
Permission	Atomic allowed action	Permission *..* Role
AcademicSession	Term-scoped governance container	Session 1..* groups, stages, calendars
Program	CS/SE/IT/AI/DS/CY/CGV or configured value	Program 1..* groups; *..* ProjectType
ProjectType	Versioned pathway and policy	Type 1..* templates; 1..* groups
ProjectGroup	Approved student team	Group 1..* memberships; 1..1 proposal baseline
GroupMembership	User-group role and dates	Links student, group, leader/member status
Proposal	Versioned project definition	Proposal 1..* reviews; belongs to group
ProposalReview	Decision/comment at workflow stage	Review *..1 proposal; *..1 reviewer
SupervisionAllocation	Supervisor assignment history	Group 1..* allocations; references faculty
Meeting	Supervisory event and signed minutes	Meeting 1..* attendance/action items
MeetingAttendance	Participant attendance state	Links meeting and user
ActionItem	Owned task with due date/status	Action *..1 meeting; *..1 owner

8. Data model overview (continued)

8.1 Key entities - delivery, evaluation, and communication

Entity	Purpose	Primary relationships
Milestone	Scheduled checkpoint	Group 1..* milestones
DeliverableTemplate	Versioned required artifact definition	Type *..* templates; milestone reference
DeliverableSubmission	Workflow submission instance	Template 1..* submissions; group owner
DocumentVersion	Immutable stored file metadata	Submission 1..* versions
AIDeclaration	Submission-level AI disclosure	Submission 1..1 declaration per version
Rubric	Versioned scoring definition	Rubric 1..* criteria; used by stages
RubricCriterion	Criterion and maximum marks	Criterion *..1 rubric
EvaluationStage	Scheduled weighted assessment	Session 1..* stages; references rubric
Evaluation	Panel assessment instance	Stage 1..* evaluations; group subject
EvaluationScore	Evaluator criterion score	Evaluation 1..* scores; criterion reference
Notification	User delivery record	User 1..* notifications; optional target link
Announcement	Targeted institutional message	Session-scoped; audience rules
AuditLog	Immutable action evidence	References actor, role, target, change

8. Data model overview - relationships and integrity

8.2 Core relationships

- An **AcademicSession** contains program participation, calendar events, role assignments, groups, catalogue versions, evaluation stages, and publication state.
- A **ProjectGroup** has time-bounded **GroupMembership** records, a versioned **Proposal**, one active primary **SupervisionAllocation**, and optional co-supervisor or industry allocations.
- The selected **ProjectType** drives suitable-program validation, type-specific **DeliverableTemplate** records, AI policy, and rubric adaptations.
- Each **DeliverableSubmission** points to immutable **DocumentVersion** records and a version-specific **AIDeclaration**; review status belongs to the submission, not the file alone.
- An **EvaluationStage** selects a versioned **Rubric**. Each **Evaluation** records panel context, and each **EvaluationScore** links one evaluator to one criterion and value.
- **AuditLog** records reference target identifiers and non-secret before/after values for consequential changes.

Integrity rules

Rule	Constraint
IR-01	A user may have only one active membership in one group per academic session unless an authorized exception is recorded.
IR-02	Only one proposal version may be the approved baseline for a group at a time.
IR-03	Only one primary supervisor allocation may be active for a group at a time.
IR-04	A submitted score must reference the rubric version active for that evaluation.
IR-05	A published result cannot reference draft or unmoderated evaluations.
IR-06	A document version and audit event are immutable after creation; corrections create a successor record.
IR-07	Historical records retain their original session, catalogue, policy, and rubric versions.

Recommended implementation note

Use surrogate immutable identifiers, explicit effective dates, unique constraints for active assignments, transactional workflow transitions, and soft retirement rather than deletion for referenced configuration.

Appendix A - Glossary

Term	Definition
Academic session	A configured FYP cycle such as Fall 2026.
At-risk group	A group matching configurable risk indicators such as overdue milestones, low verified effort, or repeated returned work.
Catalogue	Versioned set of project types, policies, outcomes, and deliverable templates.
Deliverable	A required artifact or activity evidenced through a tracked submission or completion record.
External examiner	An invited assessor with time-bounded access to assigned evaluations and evidence.
Group leader	The student member authorized to coordinate specified group actions; not an academic approver.
Industry partner	A client or mentor linked to an approved project with limited scoped access.
Moderation	Authorized review of evaluation completeness, consistency, and calculation before publication.
Project type	One of the ten framework pathways or a later approved versioned pathway.
Rubric	A versioned set of scored criteria, maxima, descriptors, and applicability rules.
Signed-off minutes	Meeting minutes reviewed and accepted by an authorized supervisor.
Verified effort	Student-reported hours confirmed or accepted through the configured supervisor process.

Appendix B - Permission matrix summary

Legend: A = perform/approve; S = submit; R = review/comment; C = contribute; V = view; - = no baseline access. A* applies only when assigned and conflict-free. This summary does not replace atomic server-side permissions.

Key action	SA	FC	HOD	Sup	Co-Sup	Stu	IE	EE	IP
Manage global configuration	A	-	-	-	-	-	-	-	-
Manage session/calendar	V	A	V	V	V	V	V	V	V
Manage catalogue/policies	V	A	V	V	V	V	V	V	V
Create/approve groups	V	A	V	V	V	S	-	-	-
Submit/revise proposal	-	V	V	R	R	A	-	-	C
Allocate supervisors	V	A	V	-	-	V	-	-	-
Manage meetings/logbook	V	V	V	A	A	A	-	-	C
Submit deliverables	-	V	V	R	R	A	-	-	C
Accept deliverables	V	V	V	A	A	V	-	-	C
Configure rubrics/panels	V	A	V	V	V	V	V	V	-
Score assigned evaluation	-	V	V	A*	A*	V	A	A	-
Moderate results	V	A	V	-	-	-	-	-	-
Publish results	V	A	V	-	-	V	-	-	-
View audit/security logs	A	V	V	-	-	-	-	-	-

Roles: SA = Super Admin; FC = FYP Coordinator; Sup = Supervisor; Co-Sup = Co-Supervisor; Stu = Student; IE = Internal Evaluator; EE = External Examiner; IP = Industry Partner.

> **Configurable baseline.** Final permissions, delegation bounds, and separation-of-duty rules require institutional approval. External access shall always be project-scoped and time-bounded.

Appendix C - Open issues requiring decision

These items are deliberately unresolved; the SRS marks related values as proposed or configurable rather than presenting them as established RSCI policy.

ID	Decision required	Proposed owner	Decision gate
OI-01	Evaluation stage and criterion weightage ratification	Academic Committee	Before evaluation setup is locked
OI-02	Group-size limits by program/session	FYP governance	Before student registration opens
OI-03	Supervisor and co-supervisor load caps	HOD / FYP Coordinator	Before allocation window
OI-04	External examiner onboarding, identity proof, and confidentiality process	Institutional governance	Before external access
OI-05	Plagiarism-tool provider, API contract, interpretation, and retention	Academic integrity / IT	Before integration testing
OI-06	Data-retention, archival, legal hold, and disposal schedule	Institutional governance	Before production go-live
OI-07	Approved hosting, storage region, backup geography, RPO, and RTO	IT / governance	Before production go-live
OI-08	Optional SSO provider and account-recovery model	IT / identity owner	Before security acceptance

Change-control rule

When an issue is decided, the owner shall record the decision, approving authority, effective session, impacted requirement IDs, configuration values, migration needs, and communication plan. Material changes to scope, grades, permissions, or policy require an SRS revision.

Appendix D - Acceptance and traceability

D.1 Acceptance strategy

Requirement class	Primary verification
Functional workflows	Role-based end-to-end scenarios with positive, negative, and state-transition checks.
Authorization	Permission-matrix tests at UI and API/object levels, including cross-session and cross-group denial.
Evaluation	Golden calculation cases, rubric-total validation, moderation/publication state tests, and correction audit checks.
Documents and declarations	Version, checksum, access, policy-match, and archive-manifest tests.
Performance	Controlled load test for 500 concurrent authenticated users and p95 interactive response target.
Security	Threat model, OWASP ASVS-aligned verification set, dependency/configuration review, and penetration test.
Availability and recovery	Monitoring evidence, backup success, and timed restore exercise against approved RPO/RTO.
Accessibility	Keyboard, focus, label, contrast, error, zoom, and assistive-technology checks against WCAG 2.1 AA target.

D.2 Traceability model

Each backlog item, design component, API endpoint, data entity, test case, defect, and release note should reference one or more stable requirement IDs. Acceptance shall not rely on module-level statements alone.

D.3 Baseline completion criteria

- All Must requirements are implemented or formally waived by the authorized governance body.
- No unresolved critical security, grade-integrity, data-loss, or authorization defect remains.
- All proposed institutional values required for production are ratified and configured.
- Operational procedures exist for provisioning, support, incident response, backup restore, archival, and result correction.
- Framework catalogue, common obligations, AI policies, and learning objectives have been reviewed against the source document.

> **End of specification.** This baseline contains 73 functional requirements across 15 modules, measurable non-functional requirements, five proposed evaluation stages, full stage rubrics, ten framework project types, shared obligations, learning objectives, a data model, and governance appendices.